Lecture-04

$$[A|I] \sim \sim\sim\sim \cdots [I|A^{-1}]$$

$$I_{3\times3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \sim \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{elementary matrix}} \quad R_2' = R_2 + 2R_1$$

How elementary matrix act

left multiplication

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}}_{R_2' = R_2 + R_1} \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{} = \underbrace{\begin{pmatrix} a & b \\ a+c & b+d \end{pmatrix}}_{R_2' = R_2 + R_1}$$

this operation reflect here. (on row)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \underbrace{\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}}_{C_1' = C_1 + C_2} = \begin{pmatrix} a+b & b \\ c+d & d \end{pmatrix}$$

Reflect on Column

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array}\right) \quad \underbrace{R_2' = R_2 - 3R_1}_{1^{st}}$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{array}\right) \underbrace{R_2' = (-1)R_2}_{2^{nd}} \sim \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & 3 & -1 \end{array}\right) \quad R_1' = R_1 - R_2$$

$R_2' = R_2 - 3R_1$ $\qquad$ $R_2' = (-1)R_2$ $\qquad$ $R_1' = R_1 - R_2$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$E_1 = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} \qquad E_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad E_3 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$\left[\underbrace{E_3 \; E_2 \; E_1 \underbrace{\begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix}}_{A}} \;\middle|\; \underbrace{E_3 \; E_2 \; E_1}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right]$$

$$\left[\underbrace{\tilde{A} \; A}_{I} \;\middle|\; \tilde{A} \right]$$

Q1:

$$\left(\begin{array}{ccc|ccc} 2 & 2 & 3 & 1 & 0 & 0 \\ \boxed{1} & -3 & 1 & 0 & 1 & 0 \\ \boxed{1} & 2 & 0 & 0 & 0 & 1 \end{array}\right) \sim \sim \cdots$$

$$\begin{pmatrix} 2 & 2 & 3 \\ 1 & -3 & 1 \\ 1 & 2 & 0 \end{pmatrix}^{-1} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \underbrace{\begin{pmatrix} -2/13 & 6/13 & 11/13 \\ 1/13 & -3/13 & 1/13 \\ 5/13 & -2/13 & -8/13 \end{pmatrix}}_{A^{-1}}$$

(b)

$$\left.\begin{aligned} 2x+2y+3z&=1 \\ x-3y+z&=1 \\ x+2y+0\cdot z&=5 \end{aligned}\right\} \longrightarrow \underbrace{\begin{pmatrix} 2 & 2 & 3 \\ 1 & -3 & 1 \\ 1 & 2 & 0 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{x} = \underbrace{\begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix}}_{b}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 & 2 & 3 \\ 1 & -3 & 1 \\ 1 & 2 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix}$$

System of equation

- Gaussian elimination (REF) + Back Substitution
- Gauss-Jordan (RREF)
- Inverse Matrix ($\det(A) \neq 0$ or Is A invertible)

Left

$$Ax = b$$

$$\underbrace{A^{-1}A}x = A^{-1}b$$

$$\underbrace{x = A^{-1}b}$$

$$Ax\,A^{-1} = \underbrace{bA^{-1}}\ \times$$

$$\underset{3\times1}{\begin{pmatrix} \\ \\ \end{pmatrix}} \quad \underset{3\times3}{\begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}}$$

Right multiplication doesn't make sense

Identity

operation: Addition

$(\mathbb{R}, +) \longrightarrow e = 0$ because $0 + a = a + 0 = a \ \forall a \in G$

$(\mathbb{R}, \times) \longrightarrow e = 1$ because $1 \cdot a = a \cdot 1 = a \ \forall a \in G$

operation: Multiplication

N.B.: Identity element depends on the binary operation defined on the set

Let's define a binary operation on $\mathbb{R}$,

$a, b \in \mathbb{R}$,

$$a * b = a + b - 1$$

Condition: e is called identity, $e * a = e * a = a$

Suppose, identity element $= e$

for Any $a \in \mathbb{R}$,

$$\underbrace{e * a = e + a - 1}$$

rules: $e * a = a$

$e + a - 1 = a \Rightarrow \boxed{e = 1}$

$a * e = a + e - 1$

rules: $a * e = a$

$a + e - 1 = a$

$\boxed{e = 1}$

$\boxed{e = 1}$

Q2

$\mathbb{R}^2$, $\begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2$, $\begin{pmatrix} c \\ d \end{pmatrix} \in \mathbb{R}^2$

Normal Addition, $\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$

or $(a, b) + (c, d) = (a+c, b+d)$

$e = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

New Addition, $\begin{pmatrix} a \\ b \end{pmatrix} \oplus \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c+2 \\ b+d-1 \end{pmatrix}$

or $(a, b) \oplus (c, d) = (a+c+2, b+d-1)$

suppose our identity is

$e = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$$e \oplus \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \oplus \begin{pmatrix} c \\ d \end{pmatrix} = \underbrace{\begin{pmatrix} \alpha+c+2 \\ \beta+d-1 \end{pmatrix}}$$

$$\underbrace{e \oplus \begin{pmatrix} c \\ d \end{pmatrix}} = \begin{pmatrix} c \\ d \end{pmatrix}$$

$$\begin{pmatrix} \alpha+c+2 \\ \beta+d-1 \end{pmatrix} = \begin{pmatrix} c \\ d \end{pmatrix}$$

$\alpha + \cancel{c} + 2 = \cancel{c} \Rightarrow \alpha = -2$

$\beta + \cancel{d} - 1 = \cancel{d} \Rightarrow \beta = 1$

$$e_2 \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

(Exercise: Verify the right identity is the same as e)